

Exercise 8 - Introduction to Bayesian Inference

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Task 1

```
set.seed(12242987)
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.2
```

1. Build a Beta prior distribution for this Binomial scenario, which encodes the information of the German study. Reweight with a factor of 1/10.

```
alpha_prior <- (1 + 4) / 10
beta_prior <- (4068 - 4 + 1) / 10

x <- seq(0, 0.01, length.out = 1000)
prior_density <- dbeta(x, alpha_prior, beta_prior)
```

2. Build the corresponding Binomial model for the number of people suffering from the disease based on the 1279 test. Obtain the theoretical posterior distribution for this scenario.

```
n_austria <- 1279
positives_austria <- 0
alpha_post <- alpha_prior + positives_austria
beta_post <- beta_prior + n_austria

x <- seq(0, 0.01, length.out = 1000)
posterior_density <- dbeta(x, alpha_post, beta_post)
```

3. Plot the posterior density and obtain the point estimators and 95% Highest posterior density interval of the prevalence of Covid19 (=proportion of inhabitants suffering from the disease).

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.2
```

```
data <- data.frame(x = x, density = posterior_density)
ggplot(data, aes(x = x, y = density)) +
  geom_line(color = "blue", size = 1.2) +
  labs(
```

```

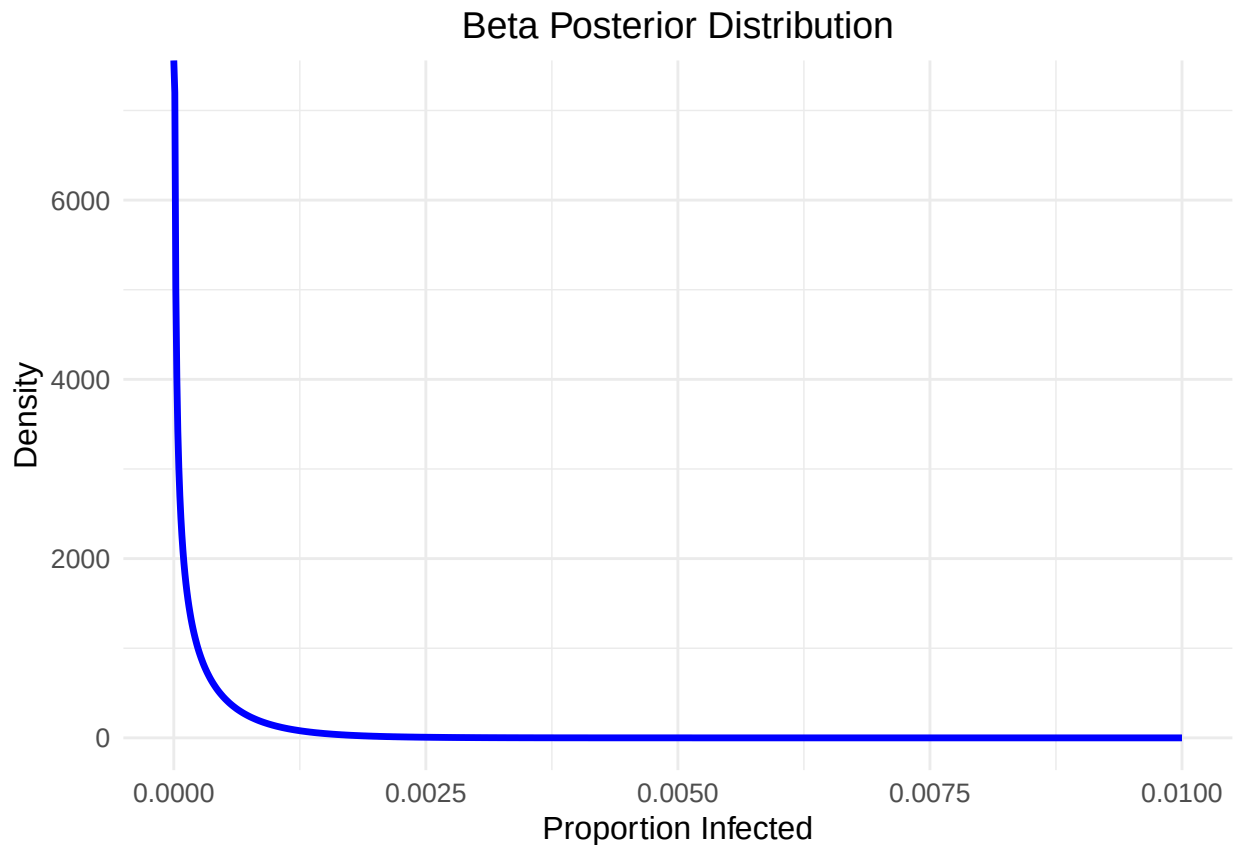
    title = "Beta Posterior Distribution",
    x = "Proportion Infected",
    y = "Density"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10)
  )

```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```



```

# point estimators
mean <- alpha_post / (alpha_post + beta_post)
median <- qbeta(0.5, alpha_post, beta_post)
mode <- if (alpha_post > 1) (alpha_post - 1) / (alpha_post + beta_post - 2) else 0

# prevalence interval
prevalence_interval <- qbeta(c(0.025, 0.975), alpha_post, beta_post)

```

Point estimators

Mean

```
mean
```

```
## [1] 0.0002965599
```

Median

```
median
```

```
## [1] 0.0001349668
```

Mode

```
mode
```

```
## [1] 0
```

95% prevalence interval

```
cat("[", prevalence_interval[1], ", ", prevalence_interval[2], "]\n")
```

```
## [ 2.913719e-07 , 0.001489436 ]
```

4. Explain why Statistik Austria chose this method instead of simulationbased or frequentist inference for obtaining intervals of the prevalence.

Statistik Austria likely chose this Bayesian inference because it incorporates prior information from the German study, and also because this makes available the point estimators and prevalence interval. The frequentist approach doesn't incorporate prior information.